EVERY OBJECT OF D_QCOH IS THE DERIVED LIMIT OF ITS TRUNCATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 680-694, immediately following the statement at 668-678.

Editorial note. The footnote (existence of a K-injective representative) is a downstream consequence, not part of the counted claim.

Allowed standard prerequisites (editorial).

- The Rlim criterion on a site with vanishing on a base. Cohomology on Sites, Lemma sites-cohomology-lemma-is-limit-dimension, cited at proof lines 692-694. Terminal input: the criterion with d = 0 for E -> Rlim tau_{>= -n}E to be an isomorphism.
- Affine vanishing of quasi-coherent cohomology. Cohomology of Schemes, Lemma coherent-lemma-quasi-coherent-affine-cohomology-zero, cited at proof lines 689-691. H^p(U, H^i) = 0 for p > 0 on affine U.

Source reference. Original reference 01XB: focused-seeds/geometry/S9-perf-nice-K-injective/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 070M: focused-seeds/geometry/S9-perf-nice-K-injective/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0H72: focused-seeds/geometry/S9-perf-nice-K-injective/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0D6P: focused-seeds/geometry/S9-perf-nice-K-injective/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 071Y: focused-seeds/geometry/S9-perf-nice-K-injective/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let E be an object of $D_{QCoh}(\mathcal{O}_X)$. Then the map $E \to R\lim \tau_{\geq -n}E$ of Derived Categories, Remark 0H72 is an isomorphism*[1].

Proof. Denote $\mathcal{H}^i = H^i(E)$ the ith cohomology sheaf of E. Let $\mathcal{B}$ be the set of affine objects of $X_{étale}$. Then $H^p(U, \mathcal{H}^i) = 0$ for all $p > 0$, all $i \in \mathbf{Z}$, and all $U \in \mathcal{B}$ as U is an affine scheme. See discussion in Cohomology of Spaces, Section 071Y and Cohomology of Schemes, Lemma 01XB. Thus the lemma follows from Cohomology on Sites, Lemma 0D6P with $d = 0$. □

[1]In particular, E has a K-injective representative, see Derived Categories, Lemma 070M.